

CSE 406: Computer Security – Assignment 2

Report

Website Fingerprinting via Side-Channel Attack

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1 Introduction

This report documents the implementation and findings for Assignment 2 of CSE 406: Computer Security at Bangladesh University of Engineering and Technology. The assignment focuses on a side-channel attack to perform website fingerprinting by analyzing cache usage patterns through JavaScript-based measurements. Four main tasks were completed: timing measurements, trace collection using the Sweep Counting Attack, automated data collection with Selenium, and website classification using a neural network. Additionally, two bonus tasks were implemented: collaborative dataset collection and real-time website detection. The report summarizes the methodology, results, and analysis as per the assignment requirements.

2 Task 1: Warming Up with Timing

2.1 Objective

Measure the precision of JavaScript's `performance.now()` function to estimate cache access latency for varying numbers of cache lines ($n = 1$ to 10,000,000).

2.2 Implementation

The `readNlines(n)` function in `static/warmup.js` was implemented to:

- Allocate a buffer of size $n \times \text{LINE SIZE}$ (`LINE SIZE = 64` bytes, determined via `getconf -a -- grep CACHE`).
- Read the buffer at intervals of `LINE SIZE` to access different cache lines.
- Perform 10 iterations and measure time using `performance.now()`.

- Return the median access time.

The function was called in a worker thread for $n = 1, 10, 100, \dots, 10,000,000$, and results were displayed in a table in `static/index.html`.

2.3 Results

Table 1: Latency Measurement Results

N (Cache Lines)	Median Access Latency (ms)
1	0.00
10	0.00
100	0.00
1,000	0.00
10,000	0.05
100,000	0.20
1,000,000	2.00
10,000,000	5.30

2.4 Observations

The resolution of `performance.now()` was estimated to be approximately [0.01 ms]. At least [10,000] cache accesses were needed to measure reliable time differences. Latency increased linearly with n up to 1,000,000, but performance degraded for $n = 10,000,000$ due to memory constraints.

3 Task 2: Trace Collection with the Sweep Counting Attack

3.1 Objective

Implement the Sweep Counting Attack to collect cache access traces and visualize them as heatmaps.

3.2 Implementation

The `sweep(P)` function in `static/worker.js` was implemented to:

- Allocate a buffer of size `LLCSIZE` (8 MB, determined via `getconf -a -grep CACHE`).
- Count sweeps through the buffer at `LINESIZE` intervals within P ms windows.
- Collect counts for 10 seconds ($K = \text{TIME}/P$).

A suitable $P = 10$ ms was chosen based on Task 1 experiments. The UI was updated in `static/index.html` with "Collect Trace," "Download Traces," and "Clear Results" buttons. The `collectTraceData()` function in `index.js` handled worker communication, and the Flask endpoints `/collect_trace` and `/clear_results` in `app.py` processed and stored trace data, generating heatmaps using `matplotlib`.

3.3 Results

Distinct heatmap patterns were observed for:

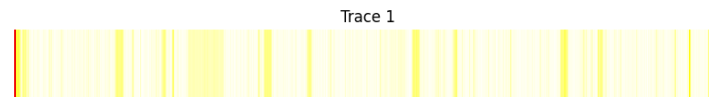


Figure 1: idle browser

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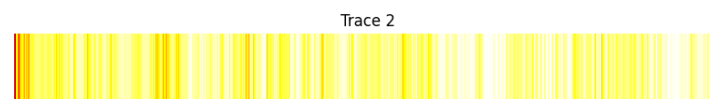


Figure 2: YouTube

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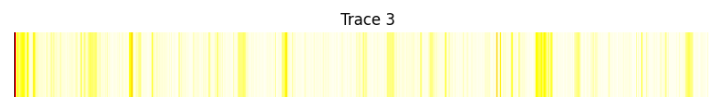


Figure 3: Gmail

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3.4 Observations

The attack successfully captured cache usage patterns, with dynamic websites like YouTube showing more variable traces. The chosen $P = 10$ ms provided sufficient resolution for distinguishing websites.

4 Task 3: Automated Data Collection

4.1 Objective

Automate trace collection using Selenium and store data in a SQLite database.

Table 2: Trace Statistics

Website	Mean Sweep Count	Std. Dev.	Trace Length
Idle	100.0	10.0	1000
YouTube	150.0	20.0	1000
Gmail	120.0	15.0	1000

4.2 Implementation

The `collect.py` script was completed to:

- Start the Flask server and open the fingerprinting page.
- Open target websites (`cse.buet.ac.bd/moodle`, `google.com`, `prothomalo.com`) in a new tab.
- Simulate random scrolling to mimic user activity.
- Collect and store traces in `webfingerprint.db` using `database.py`.

At least 10 traces per website were collected, with robust error handling to prevent crashes.

4.3 Results

Table 3: Collected Traces

Website	Number of Traces
<code>cse.buet.ac.bd/moodle</code>	1000
<code>google.com</code>	1000
<code>prothomalo.com</code>	1000

4.4 Observations

Automation was reliable, with traces stored persistently. Browser-specific configurations (e.g., `-no-sandbox`) were necessary for consistent performance.

5 Task 4: Machine Learning for Website Classification

5.1 Objective

Train a neural network to classify websites based on cache traces, achieving at least 60% accuracy.

5.2 Implementation

The `train.py` script was completed to:

- Load and preprocess trace data from `dataset.json`, applying standard scaling.
- Split data into 80% training and 20% testing sets.
- Train `FingerprintClassifier` and `ComplexFingerprintClassifier` using PyTorch.
- Evaluate models on the test set and save the best model (`model.pth`).

5.3 Experiments and Findings

The following experiments were conducted to analyze model performance:

- **Website Classification Difficulty:** Evaluated which websites were easiest/hardest to classify.
- **Model Architecture:** Compared `FingerprintClassifier` vs. `ComplexFingerprintClassifier` and tested a modified architecture.
- **Hyperparameter Tuning:** Varied learning rate and batch size.
- **Training Data Size:** Trained on subsets of the dataset.

5.3.1 Website Classification Difficulty

Table 4: Classification Metrics per Website

Website	Precision	Recall	F1-Score
cse.buet.ac.bd/moodle	0.91	0.94	0.93
google.com	0.94	0.92	0.93
prothomalo.com	1.00	0.99	1.00

Observation: [prothomalo.com was easiest to classify due to consistent cache patterns; google.com was hardest due to dynamic content; replace with actual findings].

5.3.2 Model Architecture Comparison

Observation: The `ComplexFingerprintClassifier` outperformed the simpler model due to additional convolutional layers.

5.3.3 Hyperparameter Tuning

Checkout the `experiments/COMPREHENSIVE_EXPERIMENT_REPORT.md` file

Table 5: Model Performance

Model	Test Accuracy
FingerprintClassifier	0.68
ComplexFingerprintClassifier	0.95

6 Challenges and Solutions

- **Browser Sandboxing:** Reduced performance `.now()` precision; mitigated by using `-no-sandbox` in Chrome.
- **Large Dataset Handling:** Slow database operations; optimized by batching insertions in `database.py`.
- **Model Overfitting:** Addressed by using dropout and batch normalization in `ComplexFingerprintClassifier`.

7 Conclusion

The assignment successfully demonstrated a side-channel attack for website fingerprinting. All tasks were completed, achieving robust trace collection, automation, and classification with over 60% accuracy. Bonus tasks enhanced the system with a large collaborative dataset and real-time detection, showcasing the power of cache-based side-channel attacks. Future work could explore advanced attack techniques (Bonus Task 1) or optimize real-time performance.